

## 4.8 KEYWORDS

|                             |                                                                                                                                                                                                           |
|-----------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Confidence Interval</b>  | It refers to a rule used to construct an interval. The interval keeps the estimated values over repeated samples within a margin of error limited to the level of significance (confidence level) chosen. |
| <b>Confidence Level</b>     | It refers to the percentage of samples in which the estimated sample value remains closer to the population value.                                                                                        |
| <b>Consistent Estimator</b> | An estimator that converges in probability to the population parameter as the sample size grows without bound.                                                                                            |
| <b>Heteroscedasticity</b>   | It refers to a situation where the variance of the error term is not constant.                                                                                                                            |
| <b>OLS Method</b>           | It refers to the method of estimation in which the 'sum of squared residuals' is minimum.                                                                                                                 |
| <b>R-Squared</b>            | It refers to the proportion of the total variation in the dependent variable that is explained by the independent variables included in the regression model.                                             |

## 4.9 SOME USEFUL BOOKS

- Gujarati, D N and D C Porter (2009), *Basic Econometrics*, Fifth Edition, McGraw Hill
- Johnston, J., Dinardo, J. (2007), "*Econometric Methods*", 4<sup>th</sup> Edition, McGraw-Hill Education (Asia), International Edition, pp. 109-125
- Maddala, G. S. and K. Lahiri (2009), *Introduction to Econometrics*, Fourth Edition, John Wiley and Sons
- Wooldridge, Jeffrey (2022), *Introductory Econometrics*, Seventh Edition, Cengage Learning

## 4.10 ANSWERS/ HINTS TO CHECK YOUR PROGRESS EXERCISES

### Check Your Progress 1

- 1) Go through Sub-Section 4.21 and answer.

$$2) \quad \widehat{\beta}_2 = \frac{\sum_{i=1}^n x_i y_i}{\sum_{i=1}^n x_i^2} = \frac{2179.204}{4025.43} = 0.541326$$

$$\widehat{\beta}_1 = \bar{Y} - \widehat{\beta}_2 \bar{X}$$

$$\bar{Y} = \frac{\sum_{i=1}^n Y_i}{n} = \frac{3101.35}{526} = 5.896103$$

$$\bar{X} = \frac{\sum_{i=1}^n X_i}{n} = \frac{6608.0}{526} = 12.5627$$

$$\text{Therefore, } \widehat{\beta}_1 = 5.896103 - 0.541326 (12.5627) = -0.9043$$

- 3) The computed value of  $\widehat{\beta}_2 = 0.541326$  means that for every one-year increase (decrease) in formal education of  $X_i$  (measured in years), the hourly wage rate  $Y_i$  (measured in dollars per hour) increases (decreases) by 0.5413 dollars per hour or 54.13 cents per hour.
- 4) You should describe the simple regression model and its residual in terms of equation (4.3) in the text. Subsequently, you should specify the normal probability density function of the error term and the likelihood function as equation (4.32).

### Check Your Progress 2

- 1) The variance of the OLS estimators is given in Section 4.3.
- 2) The algebraic properties of OLS estimators are described in Sub-Section 4.4.1.
- 3) The statistical properties of OLS estimators are unbiasedness, consistency and BLUE. Go through Sub-Section 4.4.2 for details.
- 4) The Gauss-Markov theorem states that given the assumptions of CLRM, the least squares estimators have minimum variance in the class of all linear unbiased estimators, i.e., they are BLUE or 'best linear unbiased estimator'.
- 3) The estimators of the homoscedastic variance of the error term  $u_i$  is given by  $\hat{\sigma}^2 = \frac{\sum \hat{u}_i^2}{n-2}$ . We have  $\sum_{i=1}^n \hat{u}_i^2 = 5980.682$ ,  $n = 526$ . Therefore,  $\hat{\sigma}^2 = \frac{5980.682}{526-2} = \frac{5980.682}{524} = 11.4135$ .

### Check Your Progress 3

$$\begin{aligned} 1) \quad r^2 &= 1 - \frac{RSS}{TSS} = 1 - \frac{\sum \hat{u}_i^2}{\sum_{i=1}^n Y_i^2} \\ &= 1 - \frac{5980.682}{7160.414} = 1 - 0.8352 = 0.1648 \end{aligned}$$

This means that 16.48 per cent of the total sample variation in  $Y_i$ , the hourly wage rate, is explained by or attributable to the variation in regressor  $X_i$ , the years of formal education.

- 2) Given that  $H_0: \beta_2 = 0$  and  $H_1: \beta_2 \neq 0$ , we need to compute the test statistic  $t = \frac{\widehat{\beta}_2 - \beta_2}{se(\widehat{\beta}_2)}$ . We also need to calculate the estimated standard error of  $\widehat{\beta}_2$  as  $\widehat{se}(\widehat{\beta}_2) = \sqrt{\widehat{var}(\widehat{\beta}_2)} = \sqrt{0.002835} = 0.0532$ . Hence,  $t = \frac{\widehat{\beta}_2 - \beta_2}{se(\widehat{\beta}_2)} = \frac{0.5413 - 0}{0.0532} = 10.1667$ . The critical value of  $t(524)$  at 5% significance level

$ist_{\alpha/2}(524) = t_{0.025}(524) = 1.960$ . Thus, at 5% significance level ( $\alpha = 0.05$ ),  $|t| = 10.167 > t_{0.025}(\infty) = 1.960$ . Therefore, we reject the null  $H_0: \beta_2 = 0$  at 5% level of significance. This indicates the existence of a relationship between the wage rate and the number of years of formal education in our sample.

- 3) The 95% confidence interval of  $Y_0$  for the given  $X_0$  is wider than the mean value  $Y_0$ . When the 95% confidence interval for each value of  $X$  is plotted, we get a confidence band. The width of the confidence band is the smallest when  $X_0 = \bar{X}$  and it gets wider as  $X_0$  moves away from  $\bar{X}$  value. Thus, the predictive capacity of  $X_0$  decreases as it departs from the  $\bar{X}$ .



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## UNIT 5 RESIDUAL ANALYSIS \*

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### Structure

- 5.0 Objectives
- 5.1 Introduction
- 5.2 Issues in Estimation
- 5.3 Analysis of Residuals
  - 5.3.1 Residual Plot
  - 5.3.2 Detection of Potential Problems
  - 5.3.3 Residuals against predicted Values
  - 5.3.4 Residual Plot over Time
- 5.4 Outliers
- 5.5 Visual Detection of Heteroscedasticity
- 5.6 Visual Detection of Autocorrelation
- 5.7 Test for Normality
- 5.8 Certain Special Cases
  - 5.8.1 Regression through the Origin
  - 5.8.2 Change in Origin and Scale of the Variables
  - 5.8.3 Linearisation of Certain Non-Linear Forms
  - 5.8.4 Choice of Functional Form
- 5.9 Limitations of Regression Analysis
- 5.10 Let Us Sum Up
- 5.11 Key Words
- 5.12 Some Useful Books/References
- 5.13 Answers/Hints to Check Your Progress Exercises

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### 5.0 OBJECTIVES

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After going through this unit, you should be in a position to

- appreciate the importance of model diagnostics in econometrics and its role in ensuring the reliability of regression analysis results;
- determine techniques for analysing residuals such as plotting residuals against predicted values and over time to detect outliers
- assess the estimated regression model in terms of goodness of fit.
- analyse visual methods for detecting heteroscedasticity and autocorrelation in regression models;
- explore testing for normality; and
- address special cases in regression analysis such as regression through the origin, variable origin and scale adjustments, and optimal functional form selection.

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## 5.1 INTRODUCTION

The pursuit of estimates for parameters that underlie economically meaningful relationships has consistently stood at the core of applied econometric modelling endeavours. Consequently, estimation of parameters is an important step in econometrics.

Within the framework of the Classical Linear Regression Model (CLRM), a pivotal assumption stipulates that the model employed for analysis is “correctly specified” (see Unit 8 for details on specification issues). This requirement presents a formidable challenge, as the notion of a flawless model remains elusive. In the realm of econometric modelling, the objective is to encapsulate the essential characteristics of an economic phenomenon while considering elements such as underlying economic theory, prior empirical research, intuition, and research expertise. Attempting to encompass every conceivable factor that impacts a particular research subject would lead to the creation of an impractical and cumbersome model with restricted practical applicability.

It is not always possible to build a perfect econometric model at the first instance. After estimation of the parameters we check whether the model satisfies the assumptions and whether the model is specified properly. We will discuss specification issues in Unit 8. In this Unit we will try to examine the model on the basis of certain rudimentary methods such as visual inspection of the residuals.

## 5.2 ISSUES IN ESTIMATION

We have studied the four assumptions regarding the error term in a simple regression model. These assumptions are:  $E(u_i) = 0$ ;  $E(u_i^2) = \sigma^2$ ;  $E(X_i u_i) = 0$ ;  $E(u_i u_j) = 0$ . These assumptions are ideal scenarios – they are not fulfilled in all the cases. In particular, when  $E(u_i^2) \neq \sigma^2$ , we encounter the problem of heteroscedasticity (i.e., the error variance is not constant throughout the dataset). When  $E(X_i u_i) \neq 0$ , we have the problem of endogeneity (i.e., the explanatory variable is not independent of the error term). When  $E(u_i u_j) \neq 0$ , we face the problem of autocorrelation in the dataset. When the assumptions are violated the OLS estimators do not fulfil the desirable properties of an estimator.

There are test procedures to detect the problems of heteroscedasticity, endogeneity and autocorrelation. However, by looking into the residual plot of a regression model, we can sense the possibility of certain problems in the dataset.

When computing the Ordinary Least Squares (OLS) estimate of the parameter ‘ $\beta$ ’, certain observations might wield disproportionate influence compared to others. Should a handful of observations exert an exceptionally strong impact, it is prudent to investigate whether they stem from erroneous data, such as decimal point misplacement, or pertain to atypical cases, such as including the CEO of Apple in your sample on monthly salary of individuals! In a broader context, it is worthwhile to assess the robustness of the estimation results concerning seemingly minor alterations in the sample. In specific instances, opting for more

robust estimation techniques may prove advisable over employing OLS. Given that a linear regression model may not always suit the data at hand, it is essential to gauge its suitability by defining residuals and scrutinizing residual plots.

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## 5.3 ANALYSIS OF RESIDUALS

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Recall that we add an error variable ( $u$ ) to the regression model, which is not measurable. Based on the estimated coefficients of the model, we obtain the residuals ( $\hat{u}$ ). The difference between the observed value of the dependent variable ( $Y$ ) and the predicted value ( $\hat{Y}$ ) is called the residual ( $\hat{u}$ ). Each data point has one residual.

Residual = (Observed value – Predicted value)

In terms of notations

$$\hat{u}_i = (Y_i - \hat{Y}_i) \quad \dots (5.1)$$

As you know from Unit 3, the error variable ( $u$ ) is assumed to have zero mean. The sum of the residuals also is equal to zero. We denote the residual as  $\hat{u}$  or  $e$ . That is,  $\sum \hat{u}_i = 0$ . You can also write it as  $\sum e = 0$ .

Residuals can be regarded as the disparities between the actual data points and corresponding model predictions. Consequently, these discrepancies serve as a measure of the unaccounted-for variability in the dependent variable, attributable to the limitations of the regression model.

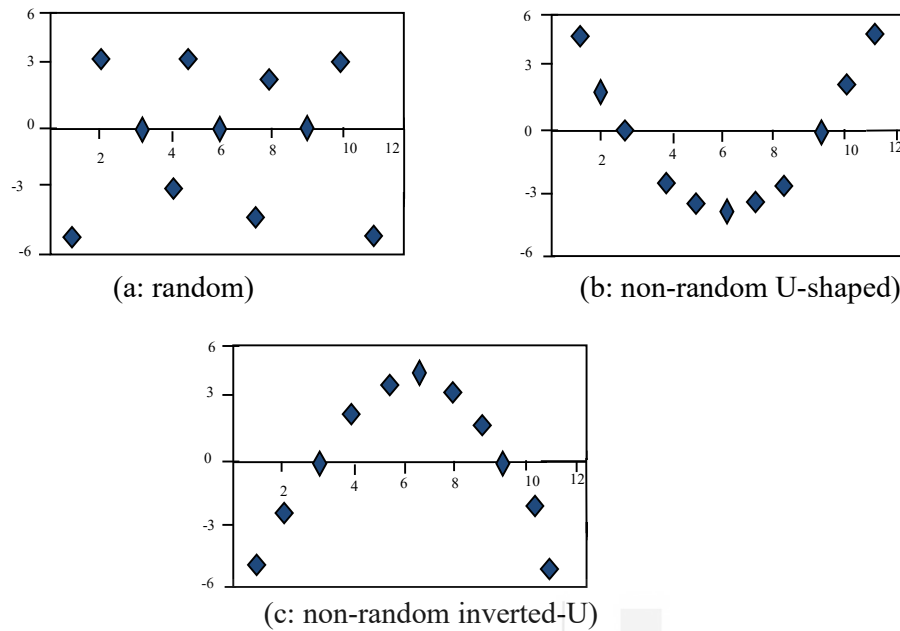
### 5.3.1 Residual Plot

A high value of the error variance ( $\sigma_u^2$ ) indicates lower explanatory power of a regression model (recall the definition of  $R^2$ ). High value of the error variance can occur if the residuals are widely dispersed. Therefore, we should find out if any of the residuals has a very high value. For this purpose, we should examine the residual plot of a regression model. First, we estimate the simple regression model ( $Y_i = \alpha + \beta X_i + u_i$ ) and then we compute the residuals.

A residual plot is a graphical representation where the horizontal axis represents the independent variable ( $X_i$ ), and the vertical axis displays the residuals ( $\hat{u}_i$ ). If the data points in a residual plot exhibit a random and evenly spread distribution around the horizontal axis, a linear regression model is a suitable choice for the data. Conversely, suppose the data points in the residual plot do not exhibit such random pattern – then it suggests that a nonlinear model would be a better alternative.

### 5.3.2 Detection of Potential Problems

Conducting a visual examination of residuals stands out as a highly efficient method for assessing the appropriateness of a regression model's fit and verifying its foundational assumptions. Different types of graphical representations are available for scrutinizing various assumptions, and these visual aids can be generated using econometric software.



**Fig. 5.1: Patterns of Residual Plot**

In Fig. 5.1 we present various patterns observed in a residual plot. There are three patterns shown in the figure. The first one (a) indicates a random pattern. The second one (b) indicates a U-shaped pattern while the third one (c) indicates an inverted-U pattern. Thus, the second and the third patterns are non-random. If we notice some non-random pattern in the residual plot, we infer that the linear regression model is not a good fit.

### 5.3.2 Residuals against Predicted Values

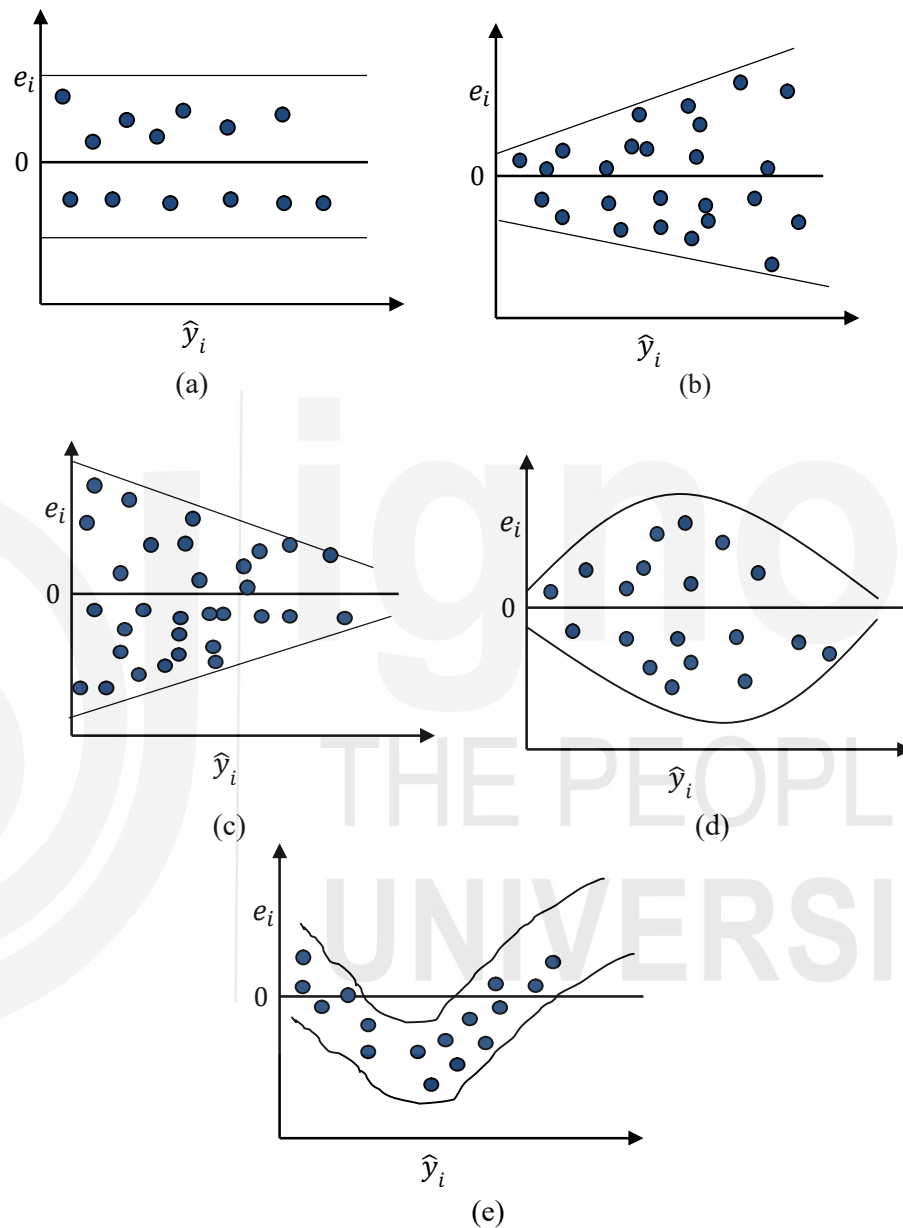
Creating a plot of residuals (e.g.,  $\hat{u}_i$  or any of the scaled residuals (see subsequent section on scaling the variables) against their corresponding predicted or fitted values ( $\hat{Y}_i$ ) proves valuable in identifying various prevalent model deficiencies. Different interpretations can be attributed to specific types of plots featuring  $\hat{Y}_i$  versus  $\hat{u}_i$ .

If the residual plot exhibits a pattern where the residuals can be enclosed within a horizontal band (see Fig. 5.2(a)), and these residuals fluctuate somewhat randomly within that band, it signifies the absence of model flaws. On the other hand, if the plot displays a pattern where the residuals are enclosed within an outward-opening funnel shape (see Fig. 5.2(b)), then the error variance is not constant. The error variance ( $\sigma_u^2$ ) increases with the predicted values ( $\hat{Y}_i$ ). When examining the plots and observing a pattern where the residuals can fit within an inward-opening funnel (see Fig. 5.2(c)), this pattern implies that the error variance is not uniform but rather decreases as a function of the predicted values ( $\hat{Y}_i$ ).

In the event that the plot exhibits a configuration where the residuals can fit within a double bow shape (see Fig. 5.2(d)), this particular pattern serves as an indicator that the error variance is not uniform but instead is related to the predicted values ( $\hat{Y}_i$ ) in a proportional manner, ranging between 0 and 1. In such cases,  $Y$  might follow a ‘binomial distribution’ and suggests that the presumed relationship between  $Y$  and the predictor variables ( $X_s$ ) is nonlinear.

When the plot reveals a pattern where the residuals are enclosed within a curved shape (see Fig. 5.2(e)), this signifies nonlinearity in the model. It suggests that

the presumed relationship between the dependent variable ( $Y$ ) and the independent variables ( $X_s$ ) is nonlinear. This type of residual plot could also imply the need for incorporating additional explanatory variables into the model. In some cases, applying transformations (such as logarithmic) to the variables could prove beneficial when addressing such nonlinearity.



**Fig. 5.2: Residual Plot**

Plotting residuals against their corresponding explanatory variables can also provide valuable insights. The interpretation of these plots aligns with that of residual-versus- $\hat{Y}_i$  plots and can be summarized as follows:

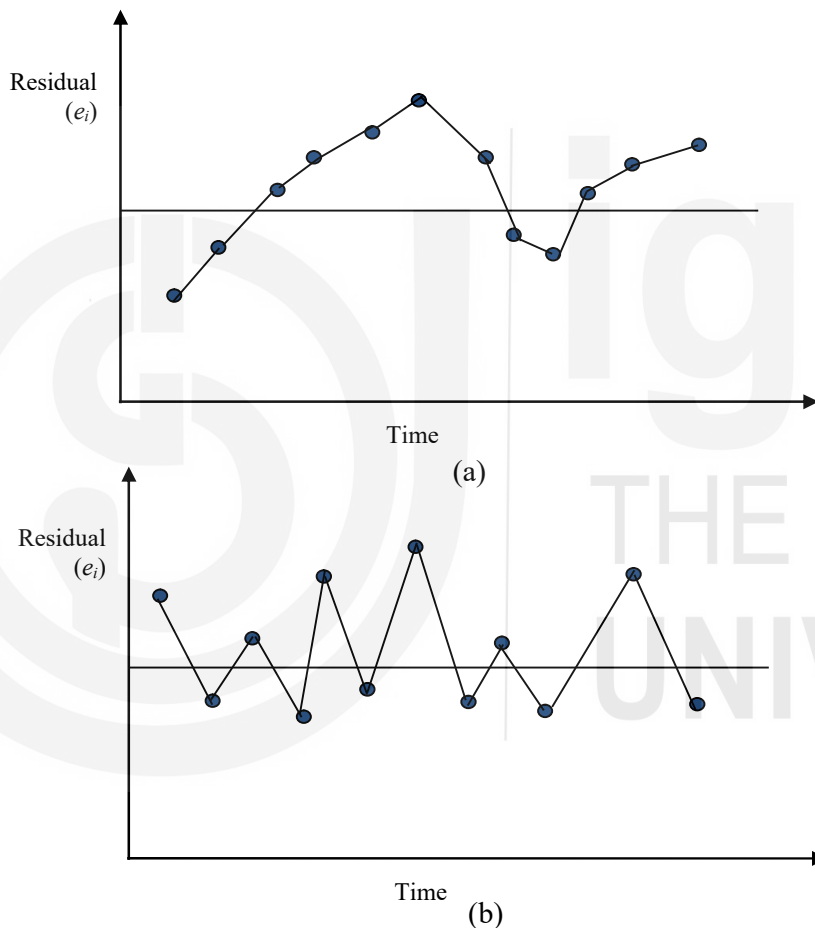
- (a) If all the residuals cluster within a horizontal band, with fluctuations appearing randomly within this band, it signifies a desirable scenario with no apparent model defects.



- (b) When the residuals take an outward-opening funnel shape or an inward-opening funnel shape, this pattern suggests that variance is not constant across the sample.
- (c) If the plot shows a double bow pattern or nonlinear trend, it implies that the assumed relationship between the dependent variable ( $Y_i$ ) and the explanatory variable ( $X_i$ ) may not be accurate. In such cases, various possibilities emerge, including  $y$  possibly being a proportion, the need for higher-order terms in  $X_i$ , or the requirement for data transformation to be considered.

### 5.3.4 Residual Plots over Time

Knowing the temporal sequence of data collection allows for the generation of residual plots with respect to time. Taking into account the Y-axis residuals and arranging them with respect to time is analogous to plotting residuals against  $\hat{Y}_i$ . In lieu of  $\hat{Y}_i$ , simply use the time order on the X-axis.



**Fig. 5.3: Inter-temporal Residual Plots**

If all the residuals are confined within a horizontal band, and they exhibit a somewhat random fluctuation within this band, it is considered favourable and suggests the absence of obvious model defects. A funnel shape that opens outward or inward suggests that the variance is not consistent but rather changing over time. If you observe a double bow pattern or a nonlinear pattern, it implies that the assumed relationship is not linear. In such instances, it is advisable to incorporate linear or quadratic time terms into the model.